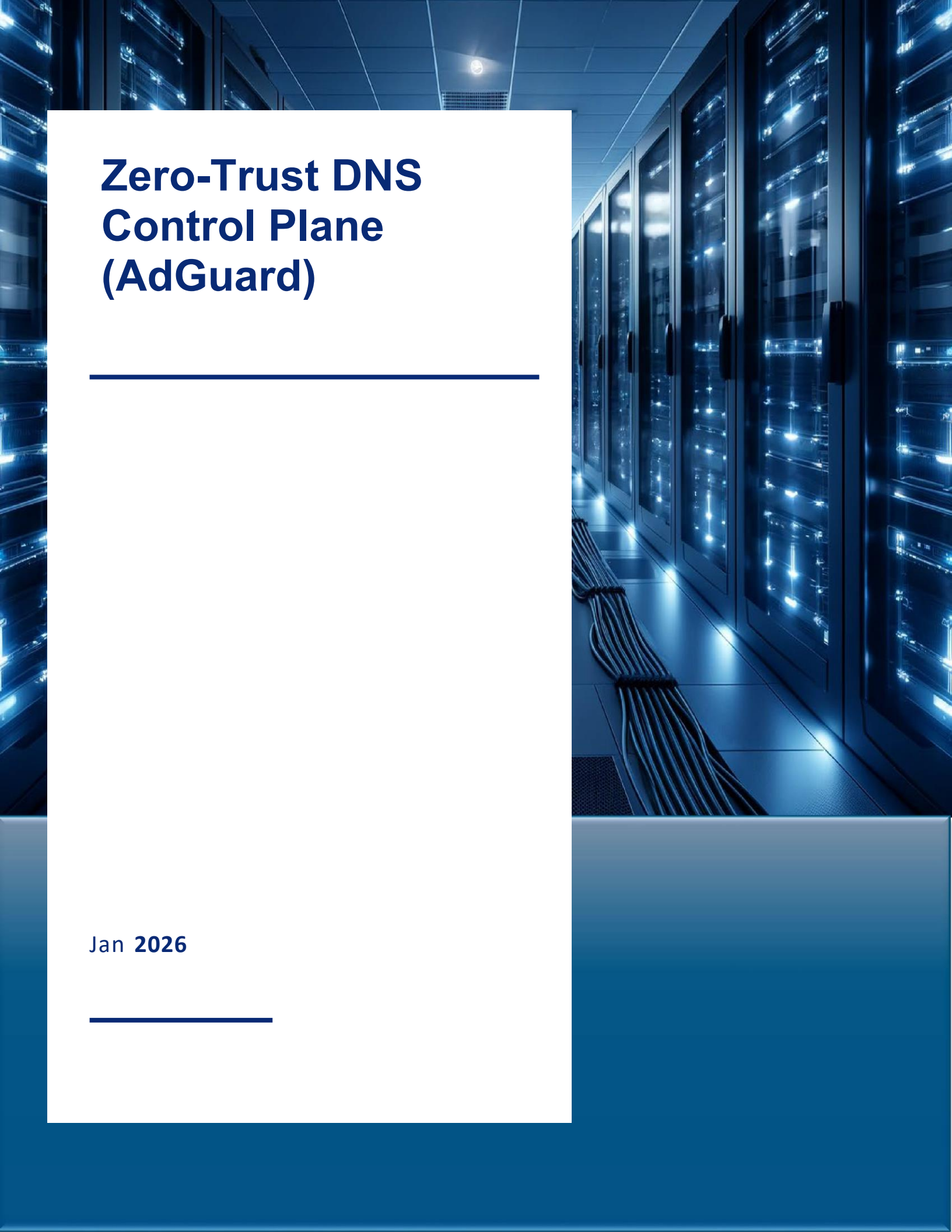




Zero-Trust DNS Control Plane (AdGuard)

Jan 2026

Introduction

Design and Scope

This project documents the design and deployment of a centralized DNS security and policy enforcement layer using AdGuard Home in a mixed-trust home network.

The environment contains personal devices, family devices, media systems, IoT cameras, smart speakers, and homelab infrastructure. DNS is used as the primary control point to manage outbound domain resolution, reduce telemetry and tracking traffic, and enforce device-specific network behavior.

Rather than applying uniform DNS filtering across all endpoints, devices are classified by role and trust level, and conditional DNS policies are applied accordingly. This enables per-device segmentation, telemetry suppression, and privileged access control using a single DNS enforcement plane.

This documentation records the architecture, policy design, and operational behavior of the deployment.

Threat Model

This deployment assumes a mixed-trust environment where not all endpoints are equally trustworthy or maintainable. The primary risks addressed are DNS-level exposure, telemetry leakage, and the inability to enforce least-privilege outbound communication across consumer and IoT devices.

In Scope

1. Telemetry and Tracking Leakage:

Consumer devices (smart TVs, mobile devices, smart speakers, wearables, cameras) routinely communicate with analytics and advertising domains. This creates privacy exposure and increases unnecessary external dependencies.

2. Malicious or Compromised Endpoint Behavior:

Any endpoint on the network may become compromised through phishing, malicious ads, vulnerable IoT firmware, or insecure applications. A compromised device can attempt to contact command-and-control infrastructure, trackers, or unknown domains. DNS is treated as an enforcement point to reduce successful outbound resolution to known abuse domains.

3. Overprivileged Network Access:

Flat networks treat all devices as peers. This increases risk when low-trust devices (IoT, media) share the same network plane as administrative and homelab systems. This deployment limits outbound resolution paths by role to reduce exposure and simplify containment.

4. Lack of Visibility and Auditability:

Without a centralized resolver, DNS activity is fragmented across devices and difficult to inspect. Centralized logging is used to observe outbound domain behavior and validate enforcement.

Threat Model

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Out of Scope

1. Telemetry and Tracking Leakage:

This deployment does not replace VLANs, firewall rules, or host-based controls. Devices on the same LAN can still potentially communicate directly by IP unless separately segmented.

2. Malicious or Compromised Endpoint Behavior:

Some clients may attempt to use DoH/DoT or hardcoded DNS. This deployment focuses on DNS policy at the resolver; preventing endpoint bypass requires additional gateway-level controls.

3. Overprivileged Network Access:

AdGuard Home can reduce successful resolution to known malicious domains, but it does not detect, clean, or remediate compromised devices. Separate endpoint security measures are required.

Security Objectives

- Reduce outbound telemetry, analytics, and advertising traffic from consumer/IoT endpoints
- Enforce least-privilege DNS behavior based on device role and trust tier
- Improve visibility into outbound domain activity through centralized logging
- Preserve functionality for administrative and homelab infrastructure while maintaining baseline protections

Zero Trust Architecture

This deployment applies Zero Trust principles at the DNS layer by treating domain resolution as a centralized policy enforcement point. Devices are not granted uniform DNS behavior simply because they are on the same LAN. Instead, DNS access is determined by device identity, role classification, and trust tier.

Architecture Summary

AdGuard Home functions as the authoritative resolver for the network and evaluates each DNS request against:

- Global filtering (blocklists and threat intelligence)
- Device and user classification (tags)
- Conditional enforcement rules (per-role allow/deny policies)

DNS responses are therefore policy-controlled rather than universally permissive.

Client Identity and Role Classification

Each endpoint is represented as an AdGuard “client” and assigned one or more tags. Tags map devices into roles such as phones, cameras, TVs, tablets, infrastructure, and administrative systems.

Policy Layer (Conditional Rules)

Custom filtering rules enforce different resolution behavior depending on client identity. This allows selective restriction of telemetry and tracking domains for specific device classes while preserving functionality for privileged administrative devices.

Trust Model

This environment uses a role and trust tier model:

- **user_admin**: administrative workstations and homelab infrastructure
- **user_regular**: normal user endpoints
- **device_tags**: device function classification (phone, tablet, camera, TV, audio, other)

Network Enforcement Design

This section documents how DNS enforcement is applied in the network, how clients are directed to AdGuard Home, and how policy decisions are executed operationally.

DNS Placement and Control Point

AdGuard Home is deployed as a dedicated service within the homelab environment and functions as the network's primary DNS resolver. All clients are configured to use AdGuard as their DNS server so that outbound domain resolution is centralized through a single enforcement point.

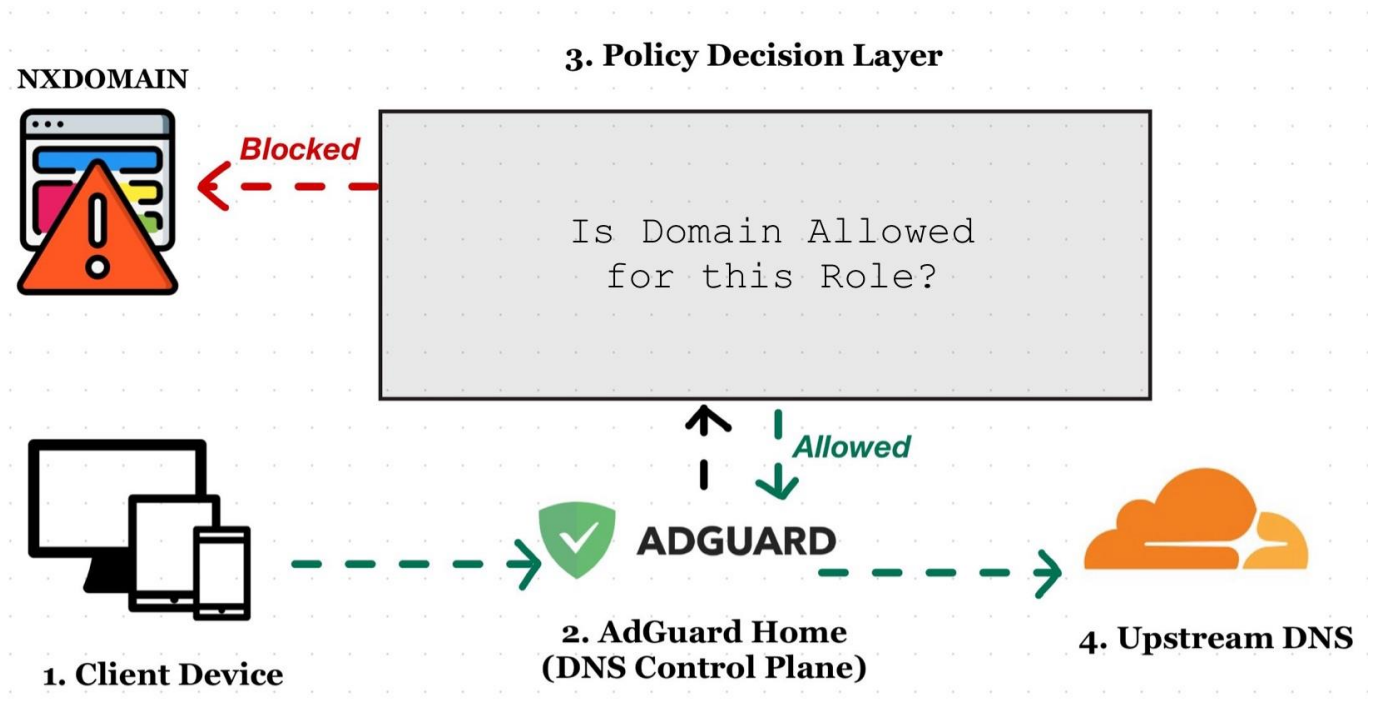
Client DNS Configuration

DNS enforcement is achieved by setting AdGuard Home as the authoritative DNS server at the network level (gateway/router DNS configuration). This ensures that:

- Existing clients automatically begin using AdGuard without per-device configuration
- New clients inherit the DNS resolver configuration upon joining the network
- DNS visibility and filtering remain centralized

Where necessary, static IP reservation is used to keep the AdGuard host address stable for the DNS configuration.

Resolution and Enforcement Flow



Device Classification Model

This section documents how endpoints are identified, grouped, and assigned trust roles within the DNS control plane.

Client Identity

Each endpoint on the network is represented as a persistent client within AdGuard Home. Clients are identified using IP address, MAC address, and DHCP reservation mapping. This ensures consistent identity and policy application even across reboots and reconnects.

Role Based Classification

Devices are grouped using client tags to define their functional role and trust tier. These tags determine which DNS policies apply to the device.

Tag	Description
user_admin	Administrative Workstations & Homelab Infrastructure
user_regular	Normal User Endpoints
device_phone	Mobile Phones
device_tablet	Tablets
device_laptop	Laptops
device_tv	Smart TV's & Streaming Devices
device_camera	Surveillance Cameras & IoT Cameras
device_audio	Smart Speakers & Voice Assistants
device_other	Wearables & Unidentified IoT Devices

Trust Model

Devices are not trusted by default. New or unclassified devices inherit only baseline DNS filtering and are treated as low trust until explicitly assigned a role. Privileged access is granted only after explicit classification.

Role-Based Policy Model

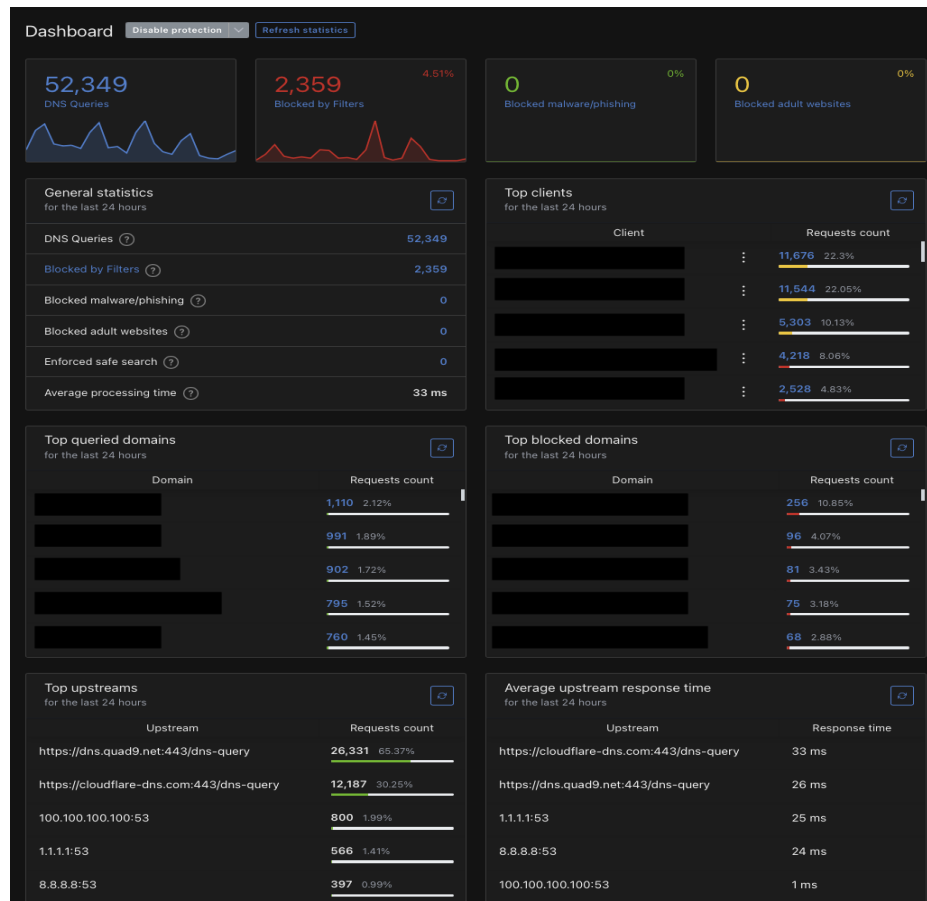
Each client tag defines a role and an associated DNS enforcement profile. Devices inherit baseline global filtering and may also be subject to additional conditional policies based on role.

Tag	DNS Behavior
user_admin	Exemptions for development, cloud, and infrastructure services. Reduced filtering
user_regular	Baseline global filtering only
device_phone	Ads, Meta tracking, and Apple telemetry endpoints restricted
device_tablet	Apple telemetry and advertising attribution endpoints restricted
device_laptop	Apple telemetry and advertising attribution endpoints restricted
device_tv	Subject to global blocklists and smart-TV telemetry filtering
device_camera	Analytics, Firebase measurement, and ad network endpoints restricted.
device_audio	Subject to global blocklists and IoT telemetry filtering
device_other	Most restrictive baseline profile. No special exemptions

Metrics

The following dashboard represents a live 24-hour operational snapshot of the DNS enforcement layer.

Live DNS Enforcement Overview



- **Total DNS Queries**
All domain lookups generated by devices on the network.
- **Blocked by Filters**
DNS requests denied by policy, including telemetry, tracking, and advertising domains.
- **Top Clients**
Devices generating the most DNS traffic.
- **Top Queried Domains**
The most commonly requested domains across all devices.
- **Top Blocked Domains**
Domains most frequently blocked due to policy restrictions and telemetry suppression.
- **Top Upstream**
External DNS resolvers used after policy evaluation is complete.
- **Average Processing Time**
Time required to evaluate and resolve each DNS request through the policy engine.